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MIL-F-16377/36(SHIPS)
30 November 1970
SUPERSEDING
MS17159(SHIPS)
29 August 1960
MS16788B(SHIPS)
5 July 1966
(See page 9)

MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTS, INCANDESCENT, GENERAL LIGHTING, 21 CANDLEPOWER OR 25 WATTS, 115 VOLTS, WATERTIGHT, SYMBOLS 95.1, 96.1, 97.1 and 98.1

The complete requirements for procuring the lights described herein shall consist of this document and the latest issue of MIL-F-16377(SHIPS).



Figure 1 - Lights, incandescent, Symbols 95.1, 96.1, 97.1, and 98.1.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

FSC 6210

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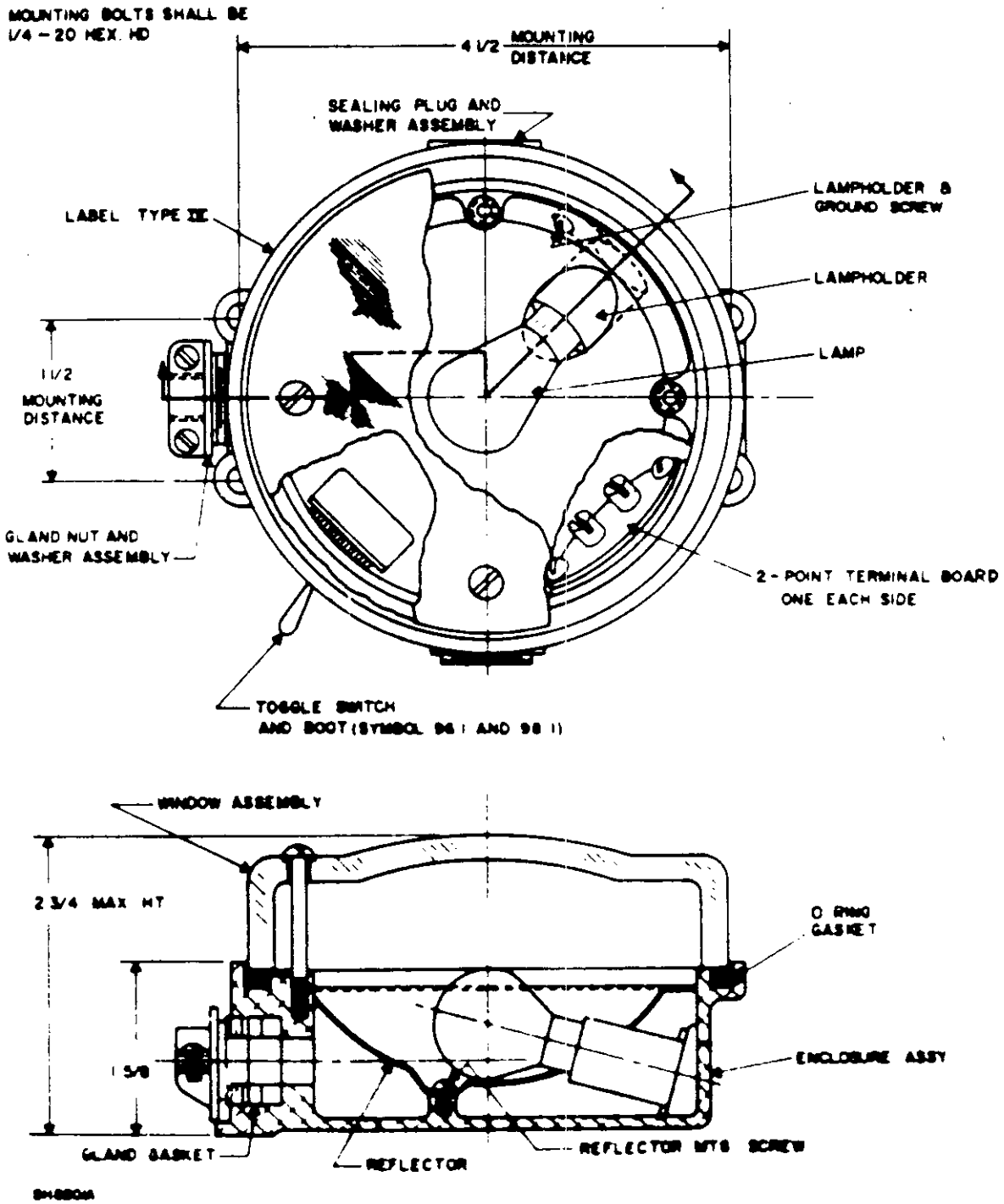


Figure 2 - Lights, incandescent, Symbols 95.1, 96.1, 97.1, and 98.1.

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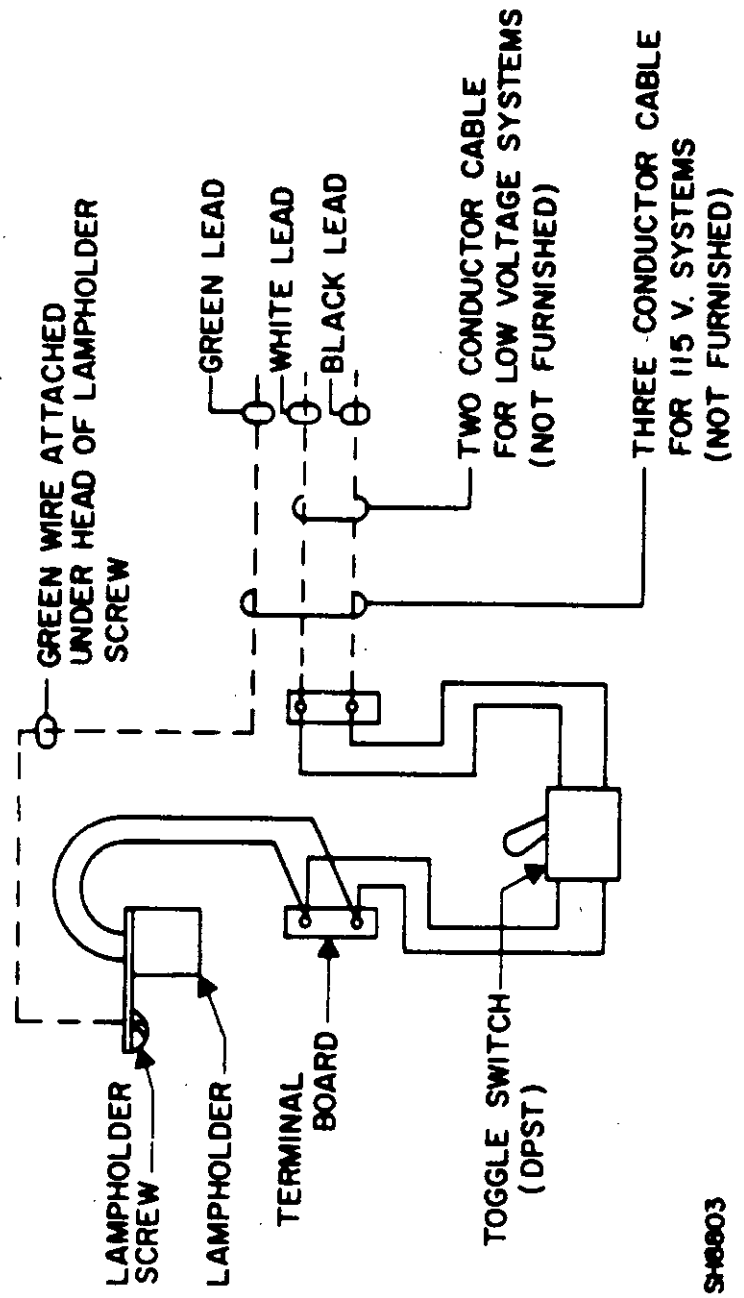


Figure 3 - Wiring diagram for symbols 96.1 and 98.1.

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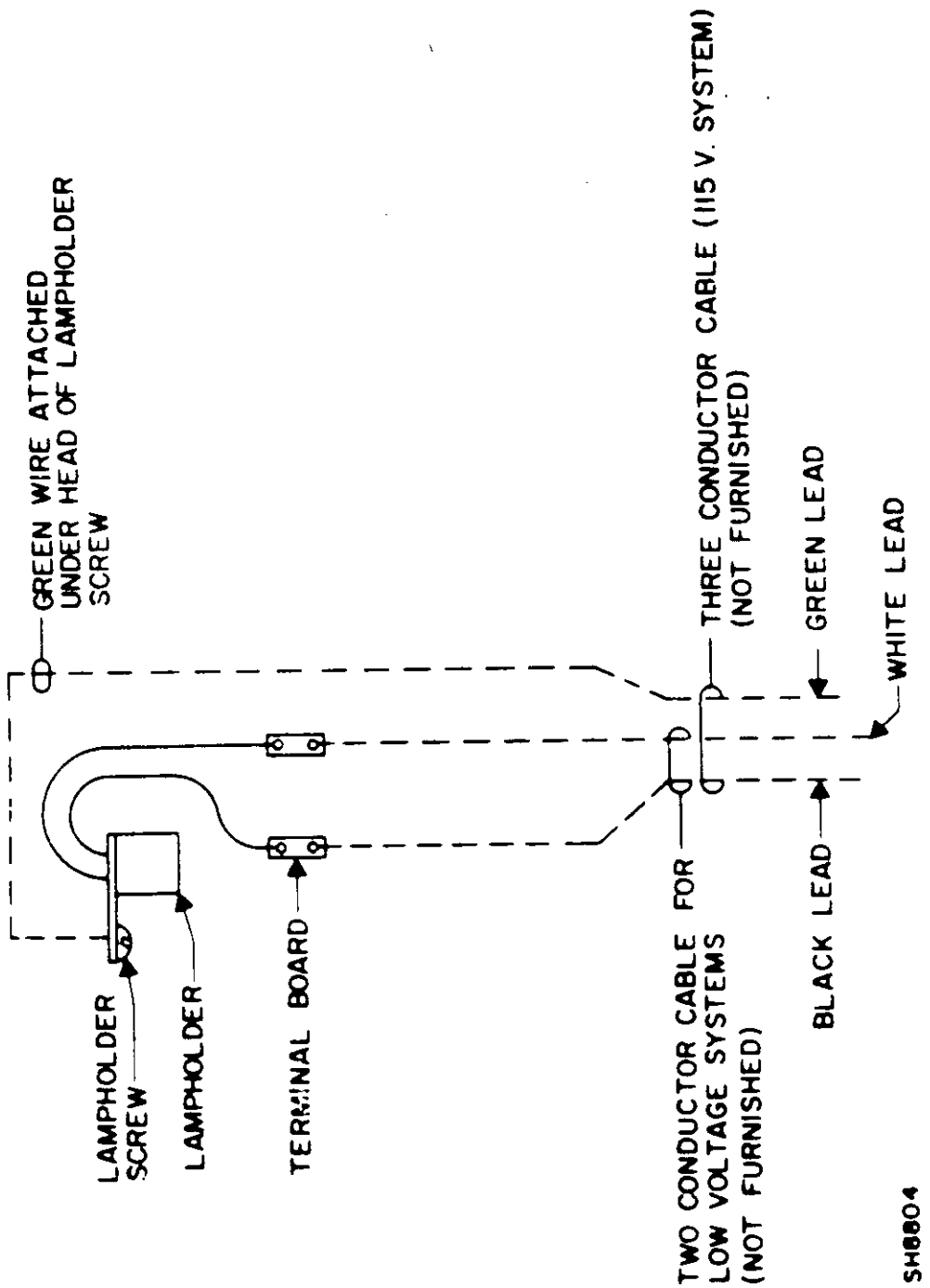
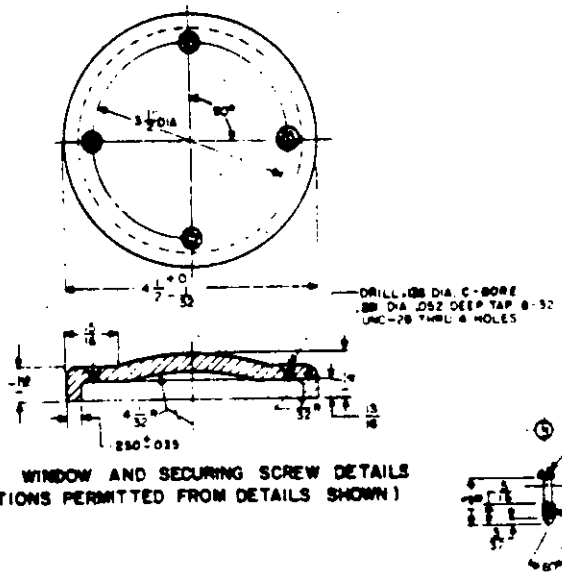


Figure 4 - Wiring diagram for Symbols 95.1 and 97.1.

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WINDOW AND SECURING SCREW DETAILS
(NO DEVIATIONS PERMITTED FROM DETAILS SHOWN)

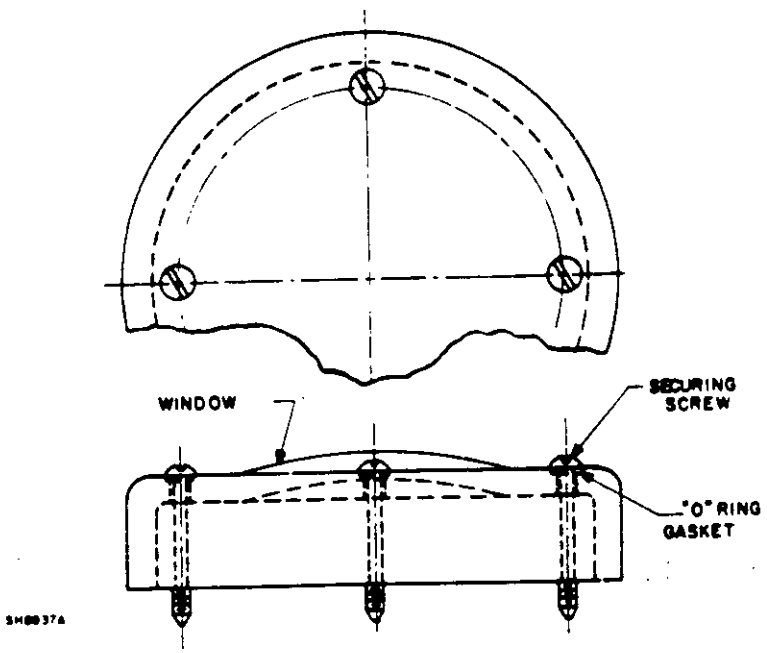
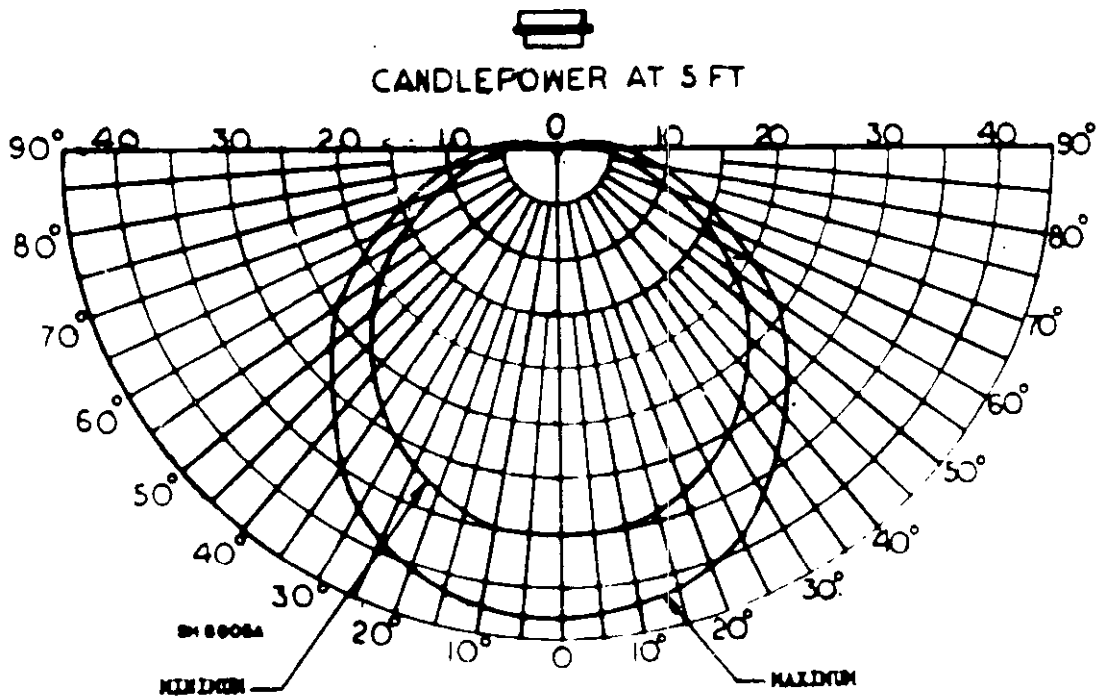


Figure 5 - Window assembly, diffusing.

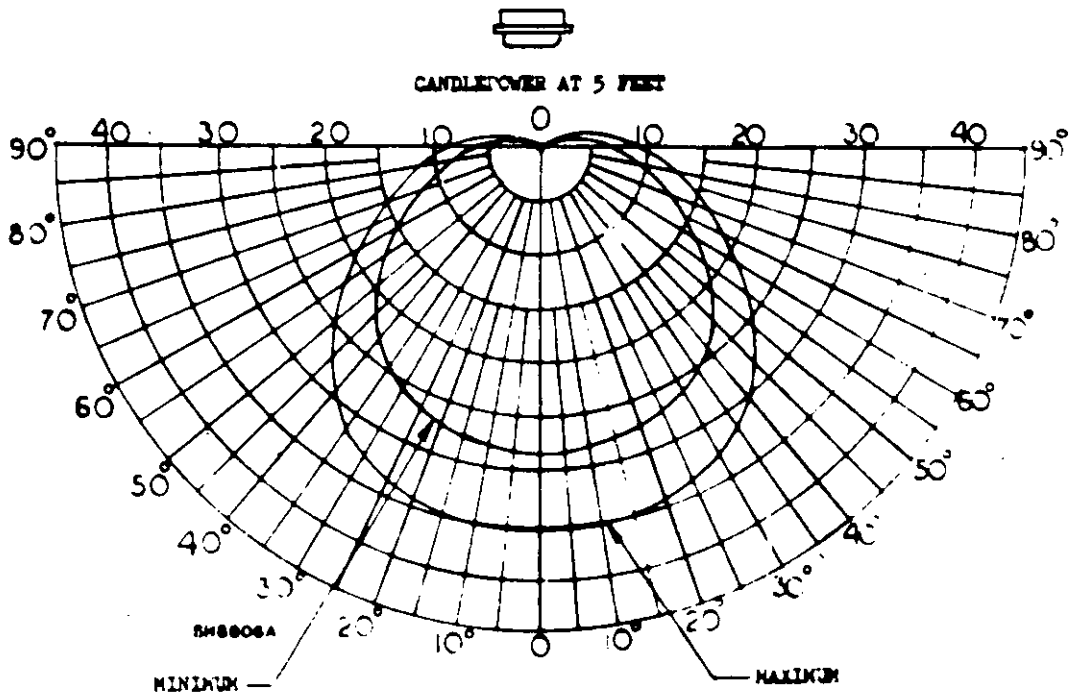
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Photometric data:
Distance: 5 feet.
Volts: 28.
Temperature 78°F.
Test shall be conducted with one 21 candlepower lamp conforming to MS15584-6.

Figure 6 - Candlepower distribution curves for symbols 95.1 and 96.1 (for 21 candlepower lamp).

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Photometric data:
Distance: 5 feet.
Volts: 115.
Temperature 78°F.
Test shall be conducted with one 25 watt lamp conforming to MS15548.

Figure 7 - Candlepower distribution curves for symbols 95.1 and 96.1 (for 25-watt lamp).

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REQUIREMENTS:
Type: II
Class: 2
Symbol 95.1: Light, with (white diffusing) window.
Symbol 96.1: Light, with (white diffusing) window and DPST switch.
Symbol 97.1: Light, with (red diffusing) window.
Symbol 98.1: Light, with (red diffusing) window and DPST switch.
Design:
Electrical rating: 115 volts, 25 watts.
Dimensions and configuration: See figures 1, 2, and 5.
Enclosure assembly: See figures 1 and 2. All metal parts shall be aluminum, brass, or steel. If steel is used, it shall be suitably protected against corrosion.
Gaskets:
O-ring: Neoprene rubber, 65-75 durometer hardness, nominal size 3/16 inch wide by 4-1/8 I.D. by 4-1/2 O.D.
Gland gasket: Neoprene sponge, 13/32 inch I.D. by 25/32 inch O.D. by 3/16 inch thick.
Window assemblies: See figure 5. Material shall be as follows:
(a) Windows:
(1) Symbols 95.1 and 96.1: White translucent, heat resistant cast acrylic plastic sheet, 1/4 inch thick, type WTLS-4, grade G of MIL-P-24191.
(2) Symbols 97.1 and 98.1: Red translucent, heat resistant cast acrylic plastic sheet, 1/4 inch thick, type RTLS-1 of MIL-P-24191.
(b) Securing screws: Brass, nickel plated or corrosion resisting steel.
(c) O-ring gaskets: AN6227-2.
(d) Coating: Anti-static coating shall be applied to all surfaces of the windows.
Reflector: Aluminum, white enamel finish.
Lampholder: MIL-L-970/15.
Terminal boards: Two point type with laminated phenolic base, commercial design.
Toggle switch for Symbols 96.1 and 98.1: Double pole, single throw, MS16569-1.
Boot for toggle switch: Neoprene rubber, with integral hexagon nut, enclosed or through type in accordance with MIL-B-5423.
Gland nut and washer assembly: One assembly shall be installed in light. Three additional assemblies shall be furnished and packaged with light to provide for additional branch lighting circuits.
Sealing plug and washer assembly: Three shall be installed in light where shown on figure 2.
Distribution curve: Candlepower (see figures 6 and 7).
Lamp: Light shall be designed to accommodate a 21-candlepower lamp conforming to part numbers -3, -4, and -6, of MS15584, and a 25-watt lamp conforming to MS15548. (Lamps are not furnished.)
Weight: 1 pound, 5 ounces, max.
Instruction sheet: Printed instructions for proper installation of cable and packing arrangements shall be furnished with each light.
Labels: Labels shall be furnished in accordance with MIL-F-16377. One type IV label shall be applied as shown on figure 2. The information to be shown on the type IV label shall be:

"USE ONLY THE FOLLOWING LAMPS
6-8 VOLTS -21 C.P. -FSN6240-155-8694.
12-16 VOLTS - 21 C.P. - FSN6240-155-8693
28 VOLTS - 21 C.P. - FSN6240-155-8725
115 VOLTS - 25 WATT - FSN6240-155-8653"

QUALITY ASSURANCE: Quality assurance shall be as specified in MIL-F-16377 and as follows:

First article inspection:
Complete fixtures: See table II
Window assemblies: See table III
Quality conformance:
Comparison inspection: Same as the first article inspection
Inspection of product for delivery:
Examination: See MIL-F-16377
Examination for preparation of delivery: See MIL-F-16377
Quality conformance tests: See MIL-F-16377

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Table I - First article and comparison inspection for complete fixtures.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.17	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Grounding circuit	3.6.13.3	4.8.11	
Optical uniformity	3.7.4.1.1	4.8.4	
Light output	3.7.1	4.8.6.1	Symbols 95.1 and 96.1 only Candlepower
Color and luminous transmittance	3.7.3.6	4.8.6.3	Symbols 97.1 and 98.1 only
Magnetic permeability	3.4.8	4.8.16	

Table II - First article and comparison inspection for window assemblies.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.4.1.1	4.8.4	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Light output	3.7.1	4.8.6.1	Symbols 95.1 and 96.1 only Candlepower
Color and luminous transmittance	3.7.3.6	4.8.6.3	Symbols 97.1 and 98.1 only

General information: (a) Assignment of Military part numbers and general information is shown in table III.

Table III - Part number and general information.

Military Part number M16377/36-	Item name	Federal Stock number (FSN)	Supersedes part number
95.1	Complete fixture, Symbol 95.1 Consisting of: Body without switch and white window	6210-295-2192 6210-548-0428	1/ MS17159-95 MS17159-99
96.1	Complete fixture, Symbol 96.1 Consisting of: Body with switch and white window	6210-295-2865 6210-548-0428	1/ MS17159-96
97.1	Complete fixture, Symbol 97.1 Consisting of: Body without switch and red window	6210-295-2192 6210-399-7028	1/ MS17159-97
98.1	Complete fixture, Symbol 98.1 Consisting of: Body with switch and red window	6210-295-2865 6210-399-7028	1/ MS17159-98
001	Body with switch	6210-295-2865	
002	Body without switch	6210-295-2192	
003	White window	6210-548-0428	MS16788-3
004	Red window	6210-399-7028	MS16788-4

1/ Formerly covered on Drawing 6210-564.1-76...

Preparing activity:
Navy - SH
(Project 6210-N392)

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